# TITLE PAGE

## Problem Statement Details

**Problem Statement ID:** SIH26176

**Problem Statement Title:**
ORCA Marine EcOsystem Reasoning with Collaborative Agents

**Theme:** Space Technology

**PS Category:** Software

**Team ID:** SIH26176

**Team Name:** codenova

**codenova**

# IDEA TITLE

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## PROPOSED SOLUTION

- Conversational marine intelligence platform powered by collaborative AI agents.
- Understands natural-language queries and automatically plans the required tasks.
- Combines satellite, ocean, weather and GIS evidence into one contextual response.

## HOW IT ADDRESSES THE PROBLEI

- Finds Potential Fishing Zones and marine conditions for a chosen location and time.
- Correlates hazards, sea state, weather and geofences before recommending action.
- Presents maps, alerts, rankings and evidence instead of isolated data values.

## INNOVATION & UNIQUENESS

- Multi-agent collaboration with autonomous task decomposition.
- Spatial-temporal reasoning across heterogeneous marine datasets.
- Explainable recommendations with evidence, confidence and safety guardrails.

# TECHNICAL APPROACH

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## TECHNOLOGIES

- Python + FastAPI backend
- LLM + agent orchestration
- GeoPandas, Shapely, Rasterio, PostGIS
- Pandas, NumPy, scikit-learn
- Streamlit / React + Leaflet or MapLibre

## METHODOLOGY & PROCESS

**1. UNDERSTAND**
Language + intent + context

→

**2. PLAN**
Decompose query into tasks

→

**3. RETRIEVE**
Marine + weather + ocean + GIS

**4. REASON**
Spatial-temporal correlation

→

**5. DECIDE**
Risk / PFZ / route ranking

→

**6. EXPLAIN**
Evidence + map + alert

**Specialist agents:** Planner • Marine Data • Weather • Ocean • Geospatial • Risk • Route • Visualization • Explanation

**Decision loop**

Retrieve evidence
Normalize data
Spatial-temporal join
Risk / route reasoning
Explain with evidence

# FEASIBILITY AND VIABILITY

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## FEASIBILITY

- Software-first prototype; no specialised hardware required.
- Modular agents can consume public-domain APIs, datasets and advisories.
- Geospatial processing can run on commodity or cloud infrastructure.
- Architecture supports phased expansion from demo to production.

## CHALLENGES / RISKS

- Different formats, spatial resolutions and update frequencies.
- Missing or delayed observations and forecast uncertainty.
- API availability, rate limits and data-access constraints.
- Marine safety decisions require careful validation.

## MITIGATION

- Timestamping, validation and source provenance.
- Confidence and uncertainty displayed with recommendations.
- Caching and fallback data sources where permitted.
- Rule-based safety guardrails with a human-readable evidence trail.

# IMPACT AND BENEFITS

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## TARGET USERS

- Fishermen and fishing communities
- Marine and ocean researchers
- Coastal authorities
- Disaster management agencies
- Maritime operators

## SOCIAL & ECONOMIC

- Earlier awareness of adverse marine conditions.
- Better-informed fishing-zone selection.
- Reduced time and fuel spent searching for suitable zones.
- Faster access to operational marine intelligence.

## ENVIRONMENTAL & GOVERNANCE

- Avoid restricted and ecologically sensitive zones.
- Support safer and more sustainable marine operations.
- Geofencing for boundary and operational awareness.
- Evidence-backed monitoring and decision support.

**Outcome: explainable marine intelligence delivered in the user's language and location context.**

# RESEARCH AND REFERENCES

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## ISRO

- Indian Space Research Organisation
- Earth Observation and space-technology context
- Official portal: isro.gov.in

## MOSDAC

- Meteorological & Oceanographic Satellite Data Archival Centre
- Satellite, weather and ocean-state data/services
- Official portal: mosdac.gov.in

## INCOIS

- Indian National Centre for Ocean Information Services
- Ocean observations, advisories and Potential Fishing Zone services
- Official portal: incois.gov.in

Research basis: official ISRO, MOSDAC and INCOIS portals + SIH 2026 Problem Statement 26176.